

Deep Learning: Notes on a Cybernetic Pigeonfight

I. Introduction: The Absurdity and Resonance of Project Orcon

Project Pigeon, later known as Project Orcon, stands as a curious and often-dismissed chapter in the history of military technology. Conceived by American behaviorist B.F. Skinner during World War II, the project aimed to develop a guided missile controlled by pigeons.¹ While seemingly absurd, the program demonstrated remarkable technical success in training pigeons to guide missiles under simulated conditions.¹ Pigeons proved capable of pecking at targets with impressive frequency and accuracy, even under simulated missile-flight speeds.¹ Yet, despite this demonstrable efficacy, Project Orcon was ultimately canceled, dismissed by the military as "eccentric and impractical," with Skinner himself lamenting that "no one would take us seriously".¹

This paradox of technical capability clashing with cultural rejection forms the central inquiry of this chapter. The dismissal of Project Orcon was not merely a functional assessment of its technical limitations but a profound cultural judgment. This historical episode serves as a compelling case study for understanding how societies frame agency, intelligence, and control through nonhuman proxies, and how these frames dictate the very legitimacy of technological systems. The incongruity between Project Orcon's operational success and its symbolic absurdity highlights a critical dimension often overlooked in technological development: the inherent entanglement of a system's functionality with its cultural and symbolic legibility.

This chapter introduces the concept of the "thick machine," a posthuman feedback system whose operational effectiveness is inextricably linked to its symbolic and cultural acceptance. Project Orcon, it will be argued, was a foundational example of such a thick machine, where the material components (pigeons, missile hardware) and the symbolic dimensions (intelligence, control, absurdity) were inseparable. The chapter aims to extend Clifford Geertz's concepts of "deep play" and "cultural systems" into the domain of cybernetic war machines, artificial intelligence (AI), and reinforcement learning. By reinterpreting Project Orcon as a symbolic performance of

rationality rather than solely a behavioral engineering experiment, this analysis will demonstrate how the pigeon, as a trained animal, became a symbolically rejected agent—a historical mirror for contemporary discomfort with nonhuman intelligence. Ultimately, this exploration traces a lineage from Skinner's pigeons to Sutton's agents, revealing that the cultural stakes of AI today are deeply entangled in histories of trust, absurdity, belief, and control. Project Orcon was a ritual of rationality, rejected not because it failed to work, but because its functional form was symbolically unacceptable, thereby anticipating the cultural politics of machine learning today.

II. Project Orcon: A Thick Description of a Cybernetic Ritual

The Pigeon-Guided Bomb: Technical Reality and Behavioral Engineering

Project Orcon's technical design was both ingenious and remarkably direct. The guidance system was housed within the nose cone of a glide bomb, such as the "Pelican" airframe.² This section contained three lenses that projected an image of the target onto a translucent screen inside a small compartment.² This screen was mounted on pivots and equipped with sensors to measure any angular movement.²

At the core of this system were one to three pigeons, each placed in its own compartment with a view of the screen.¹ These pigeons were meticulously trained through operant conditioning to recognize the target image and peck at it with their beaks.¹ Their training involved being shown an image of the target, with food pellets provided as a reward for increasingly rapid and accurate pecks.¹ Records indicate one pigeon remarkably pecked over 10,000 times in 45 minutes, demonstrating the efficacy of the conditioning.²

The mechanism for guidance was straightforward: as long as the target image remained centered on the screen, the missile would fly straight.² If the bomb veered off course, the target image would shift towards the edge of the screen. The pigeons, following the image, would peck at its new position, causing the screen to tilt on its pivots. These movements were detected by sensors, which then sent signals to the missile's control surfaces, steering the bomb back towards the target. As the bomb

corrected its course, the pigeons would continue to peck at the re-centering image, maintaining the glide path.¹ A notable feature was Skinner's implementation of a "majority rule" for multiple pigeons: if two possible targets appeared, at least two birds would agree on one, and the third would be "punished for his minority opinion" to encourage alignment with the majority.² This technical detail, where animal behavior was engineered towards a consensus, reveals a deeper human desire for unified, "rational" decision-making, even when delegated to nonhuman agents. It anticipates how contemporary AI systems are designed with alignment mechanisms, reflecting societal values of agreement and control, rather than embracing true autonomy or dissent from nonhuman intelligence.

Skinner's Vision and the Skinner Box as Ritual Enclosure

B.F. Skinner's work on Project Orcon was a direct application of his theory of operant conditioning. This theory posits that behavior is shaped by its consequences, with rewards (positive reinforcement) increasing the likelihood of a behavior being repeated, and punishments decreasing it.⁸ Skinner believed that human behavior, like animal behavior, is primarily influenced by environmental factors and the outcomes of past actions.⁹ His experiments, often conducted in what became known as the "Skinner box," involved placing animals like rats or pigeons in controlled environments where specific behaviors (e.g., pressing a lever, pecking a disc) would yield predictable rewards.⁹ These "free operants" were thus conditioned to perform complex tasks through meticulously designed schedules of reinforcement.⁹

When viewed through a Geertzian lens of cultural systems and ritual, the Skinner box transcends its identity as a mere scientific apparatus. With its controlled stimuli, precise reward schedules, and the repetitive, formalized actions it elicits, the Skinner box can be understood as a ritual enclosure. The training of the pigeons, transforming them from "natural" birds into precise, target-tracking mechanisms, becomes a highly formalized performance. This space, seemingly mundane, becomes "sacred" in the sense that it embodies a deeply held belief in the power of scientific rationality and behavioral engineering to master and transform nature. The pigeon's pecking, though a conditioned response, is a symbolic act within this ritual, demonstrating human mastery and the potential for perfect control over living systems. This performance of control, achieved through meticulous conditioning, was intended to legitimize the technology by demonstrating its predictability and reliability.

The Stakes of the Pigeonfight

While the immediate objective of Project Orcon was to provide a jam-proof guidance system for missiles during wartime ¹, its ultimate rejection reveals stakes far beyond mere military utility. The National Defense Research Committee, despite contributing \$25,000 to the research, viewed the idea as "very eccentric and impractical".¹ Skinner himself voiced frustration, stating that "our problem was no one would take us seriously".² The program was canceled in 1944, with the official reason being that it would "seriously delay others which in the minds of the Division have more immediate promise of combat application".² This suggests a prioritization of more conventional and seemingly "serious" military projects.

The cultural and symbolic dimensions of this rejection are particularly telling. During a 1944 demonstration, despite a "perfect performance" by the pigeon, Skinner observed that the spectacle of a living bird guiding a missile "simply reminded the committee of how utterly fantastic our proposal was".¹¹ The "merriment, though restrained," was evident, indicating that the project's case was lost not due to technical failure, but due to its symbolic incongruity with the prevailing military aesthetic.¹¹ The visual of a pigeon, a common and often dismissed animal, performing a critical military function undermined the perceived seriousness and rationality of the endeavor. This highlights that the adoption of technology within a military context is not solely about efficacy but also about a performative display of rationality and gravity. The pigeon, as a "lowly" animal, disrupted the established "sociotechnical imaginaries" of what constitutes legitimate and serious military technology.¹² Military culture, shaped by beliefs about "warfighting organization" and "warrior ethos" ¹⁵, found the pigeon-guided bomb to be a violation of its aesthetic and symbolic standards of control and seriousness. The cultural rejection of Orcon was thus a ritualistic act of purification, maintaining the integrity of the military's self-conception and its vision of acceptable warfare technology.

III. Deep Play, Delegated Agency, and Symbolic Refusal

Geertz's Deep Play and Cultural Systems

Clifford Geertz's seminal work, "Deep Play: Notes on the Balinese Cockfight," provides a powerful framework for understanding Project Orcon. Geertz advocated for "thick description," a qualitative research method that goes beyond surface-level observations to interpret the context and subjective meanings of human social actions.¹⁹ For Geertz, culture is a system of inherited conceptions expressed in symbolic forms, influencing how individuals understand their world.¹⁹

The Balinese cockfight, as analyzed by Geertz, is a prime example of "deep play"—a ritual where the stakes are so high that, from a purely rational economic perspective, no one should engage in it.²¹ Yet, it is deeply meaningful within Balinese culture. The cocks are not just animals but "metaphysical projections" of the men's masculinity and status.²² Through grooming and fighting their cocks, men project their pride and emotions, and the outcome of the fight, though not leading to permanent status changes, metaphorically expresses significant life events in bursts of emotion.²² The cockfight thus serves as a "simulation of social matrix," revealing deeper social structures and cultural values, and acting as an instrument of self-analysis for the Balinese.²³

Applying this framework to Project Orcon, the "pigeonfight" can also be seen as a form of "deep play." Its "stakes" were not merely military efficacy but the symbolic legibility of human control and rationality in warfare. The "irrational trust" and "opaque optimization" associated with contemporary deep learning systems find a historical parallel in the "irrational stakes" of the cockfight. The military's rejection of Orcon signifies a refusal to accept this particular "play" as legitimate within their cultural system, demonstrating that the cultural system itself dictates what is considered rational or absurd. This highlights that the performance of control, whether in a cockfight or a cybernetic system, is always embedded within and judged by a cultural system, which determines its meaning and acceptance.

Project Orcon as Delegated Agency and Symbolic Performance

The pigeon in Project Orcon was intended to be a tool, a biological sensor and

actuator in a guidance system. However, its role extended beyond mere instrumentality; the pigeon functioned as a proxy for human decision-making in a lethal context. This delegation of agency, though framed in behaviorist terms, implicitly raised questions about who or what was truly making the decision to kill. The very act of outsourcing such a critical function to an animal, even a highly trained one, introduced an unsettling dimension.

A telling detail emerges from the historical accounts: it was hinted that a "real reason for the cessation of the project was that administrators could not come to terms with the issues surrounding the fact that pigeons had to die for the cause".¹ This suggests an unarticulated ethical boundary, a discomfort with the moral implications of sacrificing animals in such a direct, instrumental way for warfare. This discomfort points to a disavowal of responsibility, where the human decision to delegate lethal action to an animal collided with an implicit moral burden.

This historical moment serves as a powerful mirror for contemporary anxieties surrounding nonhuman intelligence, particularly in autonomous weapon systems (AWS). Ethical discussions around AWS today stress the importance of "retaining human agency and intent in decisions to use force" and emphasize that "moral responsibility and accountability" cannot be transferred to inanimate machines or computer algorithms.²⁴ The ethical concerns surrounding AWS today directly echo the implicit ethical discomfort with Project Orcon. The unsettling nature of delegating life-and-death decisions to a non-human entity, whether animal or algorithm, creates an "ethical uncanny" that drives rejection, regardless of technical efficacy. This anticipates the "uncanny valley" phenomenon in AI, where systems that are too human-like, yet not quite, elicit revulsion. The pigeon, in its very biological presence within a war machine, crossed a similar threshold of symbolic discomfort.

The Cultural Rejection of Orcon

Project Orcon's classification as "crackpot" despite its technical successes is a critical point of analysis. As Skinner observed, the military committee's "merriment, though restrained," during the demonstration indicated that the proposal was perceived as "utterly fantastic".¹¹ The military's rejection was not simply about technical limitations; it was deeply rooted in a clash with its established aesthetic of control and seriousness.

Military culture, as a distinct social system, is profoundly shaped by its "sociotechnical imaginaries"—collective visions of what is achievable through scientific innovation and social changes, often tied to ideologies and ambitions.¹² These imaginaries dictate what technologies are deemed legitimate, serious, and worthy of investment.¹⁵ A pigeon-guided bomb, despite its functional promise, violated these deeply ingrained cultural norms. The "spectacle of a living pigeon" ¹¹ undermined the scientific credibility and the performative display of rationality expected of military technology. The absurdity of using pigeons for such a high-stakes endeavor clashed with the desired image of precision, control, and human intellectual superiority in warfare.

Project Orcon, therefore, failed not in its function, but in its *signification*. The pigeon, as an agent, did not embody the "right kind of intelligence" or the appropriate symbolic gravitas required for military acceptance. This demonstrates that technological legitimacy is not solely an objective assessment of performance but a performative act deeply embedded in cultural aesthetics and symbolic coherence. The cultural rejection of Orcon was a ritualistic act of purification, maintaining the integrity of the military's self-conception and its sociotechnical imaginary of acceptable warfare technology. This pattern is echoed in current AI debates, where distrust often stems from symbolic rather than purely technical issues, highlighting the enduring power of symbolic coherence in technoscience.

Table 1: Project Orcon vs. Balinese Cockfight vs. Deep Reinforcement Learning: A Comparative Analysis of "Deep Play"

Category	Geertz's Deep Play (Balinese Cockfight)	Project Orcon (Pigeonfight)	Deep Reinforcement Learning (AI Agent)
Arrival & Outsider Status	Geertz's initial rejection by villagers, eventual acceptance ²¹	Skinner's "eccentric" idea, military skepticism ¹	Public perception of AI (hype vs. skepticism, "uncanny valley" of intelligence) ²⁸
Description of "Play"	Public event, cocks fighting to the death ²¹	Pigeons pecking at target images in a nose cone ¹	RL agent interacting with environment, maximizing reward ³⁰
Proxies/Agents	Cocks as metaphysical projection of men's	Pigeons as "pilots" / "organic control" ¹	Software agents / algorithms ³¹

	masculinity/status ²²		
Stakes	Status, pride, honor (not just money) ²²	Symbolic control, wartime precision, military legitimacy ³	Trust, accountability, control, ethical alignment ²⁴
"Deep Play" / "Deep Learning"	Irrational stakes, cultural meaning ²¹	"Irrational trust" in nonhuman agency, opaque optimization	"Irrational trust" in nonhuman agency, opaque optimization
Rejection/Acceptance	Cockfight accepted as meaningful ritual ²¹	Orcon rejected as "crackpot" despite technical success ¹	AI acceptance/rejection based on perceived intelligence, trust, and control ²⁸

IV. From Cybernetic Pigeons to Silicon Agents: The Thick Machine in the Age of AI

Cybernetics and Feedback Loops

Project Orcon fundamentally operated as a first-order cybernetic system. As defined by Norbert Wiener, first-order cybernetics focuses on a closed signaling loop where a system's action generates a change, which then provides feedback to the system, leading to adjustments aimed at achieving a desired goal.³⁸ In Orcon, the pigeons' pecking at the target image generated error signals that directly corrected the missile's flight path, maintaining it on course.¹ This continuous loop of sensing, acting, and adjusting perfectly models a first-order cybernetic system.

Skinner, while rooted in behaviorism, implicitly embedded symbolic meaning within this feedback loop. The "target" was not merely a visual stimulus but represented a military objective—a ship or a street intersection.⁵ The pigeon's conditioned response, while seemingly mechanical, was directed towards a human-defined, symbolically charged goal. This demonstrates that even in ostensibly "objective" behavioral engineering, symbolic meaning can be deeply integrated into the functionality of a

feedback system.

The connection between Skinner's behaviorism and cybernetics is notable. Cybernetics, as a precursor to AI, was "profoundly behaviorist" in its commitment to ideas of feedback, regulation, and control.³⁹ Both fields sought to understand and control systems through observable inputs and outputs, often sidestepping the "internal states" that cognitive science would later emphasize.³⁹ However, the cultural rejection of Project Orcon, and the ethical questions it implicitly raised (e.g., who is responsible for the pigeon's fate?), hint at the limitations of a purely first-order cybernetic view. These questions point towards the need for second-order cybernetics, which explicitly incorporates the observer into the system and emphasizes reflexivity.³⁸ The failure of Orcon, therefore, was not just a technical failure of a first-order system, but a symbolic failure of a nascent second-order system where human values and perceptions (the observer's perspective) disrupted the observed control loop. This illustrates that the perceived "detachment" of first-order cybernetics cannot fully account for the cultural and ethical "noise" that ultimately determines a technology's fate.

Deep Reinforcement Learning as Formalized Behaviorism

Deep reinforcement learning (RL) represents a powerful modern formalization of Skinnerian behaviorism. In RL, an "agent" learns to make decisions by interacting with an "environment" and receiving a "reward signal" for desirable actions.³⁰ The agent's goal is to maximize the cumulative reward it receives over the long run, a concept formalized as the "reward hypothesis".³⁰ This trial-and-error learning, driven by feedback, directly mirrors Skinner's principles of operant conditioning, where behaviors are shaped through rewards and punishments.³¹

In this light, modern RL agents can be viewed as sophisticated "silicon pigeons." Both learn to perform goal-directed behaviors through iterative refinement based on reinforcement. However, significant advancements have been made in this translation. While Project Orcon's pigeons relied on a direct stimulus-response mechanism, reacting to what they saw without modeling future states, modern RL agents operate within a Markov Decision Process (MDP).³² This allows RL agents to assess their current state, predict future states, and choose actions accordingly, enabling predictive modeling and autonomous exploration—a critical gain over the pigeons'

capabilities.³²

What is "lost" in this translation from biological to silicon agents is the tangible, visceral presence of a living agent, replaced by opaque algorithms. While RL formalizes Skinnerian principles, the shift from a biological "black box" (the pigeon) to an algorithmic one (the neural network) introduces a new form of opacity. Pigeons were rejected partly because their "intelligence" was too visible and yet too alien. RL agents, while technically more capable, present a different challenge: their decision-making processes are often opaque and difficult for humans to interpret, leading to what can be described as "irrational trust" or profound distrust.³⁵ This highlights that the

form of intelligence, whether biological or algorithmic, transparent or opaque, significantly impacts human acceptance and trust. This creates an "algorithmic uncanny" that directly parallels the "pigeon uncanny" of Project Orcon. The exploration-exploitation dilemma, a core challenge in modern RL where agents must balance trying new actions with leveraging successful ones³², also echoes the historical challenges of shaping behavior, demonstrating a continuous thread in the optimization of learning across different instantiations of the "thick machine."

Cultural Systems, Legitimacy, and Trust in AI

The stark contrast between the Balinese cockfight, accepted as a meaningful ritual, and the pigeonfight, dismissed as absurd, underscores the profound influence of cultural systems on technological legitimacy. Science and Technology Studies (STS) provides crucial insights here, demonstrating that scientific facts and technological systems are products of socially conditioned investigations, and that legitimacy is actively produced and can be refused.¹⁴ Project Orcon violated the established structures of legitimacy for military technology, not due to inherent technical failure, but because it clashed with the prevailing "sociotechnical imaginaries" of warfare.

Sociotechnical imaginaries are collective visions of achievable futures that combine scientific innovation with social changes, often imbued with a nation's ideologies and ambitions.¹² The military's imaginary of control and seriousness in warfare could not accommodate the image of pigeons guiding bombs. The "absurdity" of the pigeon, as perceived by military officials, violated the aesthetic and symbolic standards of control and seriousness they sought to uphold.¹⁵ This demonstrates that the symbolic

incoherence of a technology, regardless of its functional efficacy, can lead to its rejection, affirming that technoscience is always a "thick machine" where meaning and function are inseparable.

This historical pattern resonates deeply with contemporary debates surrounding trust in AI. How we treat RL agents today, whether with seriousness or dismissiveness, is influenced by an "anthropomorphism bias".²⁸ Humans tend to attribute human-like qualities to non-human entities, and this bias significantly influences the perceived intelligence and trustworthiness of AI.²⁸ When AI models generate outputs that mimic human communication, nuance, and intent, they are often perceived as more trustworthy and useful.³⁵ Conversely, when AI exhibits behaviors that are opaque, unexplainable, or deviate from human expectations, it can lead to distrust and "symbolic refusal".³⁵ The mistrust of AI, particularly in sensitive applications, is a cultural echo of the pigeon's rejection. Both cases reveal a persistent discomfort with delegating intelligence and control to nonhuman proxies, especially when these proxies defy established symbolic norms of what constitutes "acceptable" intelligence or agency.

Table 2: Key Theoretical Frameworks and Their Application to Project Orcon/AI

Framework	Core Concepts	Application to Orcon/AI
Clifford Geertz (Thick Description, Deep Play, Cultural Systems)	Cultural interpretation, symbolic meaning, ritual, irrational stakes ¹⁹	Analyzing Orcon as a symbolic ritual of rationality; comparing its stakes to deep play; understanding its cultural rejection.
B.F. Skinner (Operant Conditioning, Behaviorist Epistemology)	Stimulus-response, reinforcement, environmental control, shaping behavior ⁸	Explaining pigeon training; linking Skinner box to ritual enclosure; formalizing behaviorism in RL.
Gregory Bateson / Norbert Wiener (Cybernetics, Feedback)	Feedback loops, control, communication, observer's role ³⁸	Modeling Orcon as a first-order cybernetic system; discussing the shift to second-order cybernetics in AI.
Rich Sutton (Deep Reinforcement Learning)	Reward hypothesis, maximizing cumulative reward, MDPs,	Connecting RL agents to "silicon pigeons"; analyzing gains/losses in translation;

	exploration-exploitation ³⁰	understanding opaque optimization.
Donna Haraway / Hayles (Posthuman Agency, Companion Species)	Co-constitution, naturecultures, significant otherness, challenging human-animal binary ⁴⁴	Reinterpreting the pigeon as companion agent, not tool; exploring delegated agency and ethical implications; challenging human-tool binary.
Science & Technology Studies (STS) (Legitimacy, Cultural Resistance, Technopolitics)	Social construction of technology, sociotechnical imaginaries, legitimacy, power dynamics ¹²	Examining why Orcon was rejected; analyzing military "sociotechnical imaginaries"; understanding public trust/distrust in AI.

V. Posthuman Agency, Ethics, and the Epistemology of Delegation

The Pigeon as Companion Species

Donna Haraway's "Companion Species Manifesto" offers a compelling lens through which to re-evaluate the pigeon's role in Project Orcon. Haraway challenges the rigid human-animal binary, arguing that humans and other species are "co-constitutive" and intertwined in complex "naturecultures".⁴⁴ She posits that dogs, for instance, are not merely tools or theoretical constructs but "fleshly material-semiotic presences in the body of technoscience".⁴⁵

Applying this framework, the pigeon in Project Orcon, despite being treated as a mere instrument, can be reframed as a co-constitutive partner in a human-animal-machine assemblage. The meticulous training, the shared environment within the missile nose cone, and the reliance on the pigeon's biological capabilities for a human military objective suggest a relationship more complex than simple tool-use. Haraway's concept of "ruff love" and "disciplined spontaneity" in training ⁴⁶ offers a way to understand the human-pigeon "training stories" within Orcon: a demanding, yet deeply intertwined, process that shaped both the bird's behavior and the human

perception of its capabilities. The "implosion of nature and culture" ⁴⁵ within the pigeon-guided bomb makes it a quintessential "thick machine," where biological and technological elements merged to create a new form of agency. The military's rejection of Orcon, therefore, can be seen as a refusal to acknowledge this "significant otherness" and the inherent "impurity" of the human-animal-machine assemblage. This suggests that the denial of co-constitution and the insistence on a clear human-tool hierarchy contributed to the symbolic rejection of the "thick machine" that Orcon truly was.

Ethics of Delegated Agency and Trust

The ethical implications of Project Orcon extend far beyond its technical viability, resonating with contemporary debates on autonomous systems. The act of trusting a pigeon with a bomb, and the subsequent disavowal of responsibility for the pigeons' inevitable deaths ¹, foreshadows the complex ethical landscape of delegating lethal decision-making to nonhuman entities. The International Committee of the Red Cross (ICRC) vehemently argues against ceding "life-and-death decisions" to machines, emphasizing that "moral responsibility and accountability" are human responsibilities that "cannot be transferred to inanimate machines, or computer algorithms".²⁴ The discomfort hinted at by administrators regarding the pigeons' sacrifice suggests an early, unarticulated "moral economy" where the cost of delegating lethality to certain non-human agents (animals) was deemed too high, not in monetary terms, but in terms of symbolic and ethical burden. This unacceptable moral burden, combined with the symbolic incongruity, contributed to Orcon's rejection, foreshadowing the ongoing ethical debates about AI accountability and the "human in the loop" in autonomous systems.

The rejection of Orcon, and by extension, modern skepticism of AI, can be understood through the concept of "symbolic violence." Coined by Pierre Bourdieu, symbolic violence describes the subtle, often unconscious, imposition of norms by dominant groups onto subordinate ones, leading to their devaluation or exclusion.⁴² In this context, human society, as the dominant group, imposed its norms of what constitutes "intelligence" or "acceptable agency" onto the pigeons (and later, AI). The pigeon, despite its functional success, was symbolically "sacrificed" to uphold a human-centric vision of warfare and control. This mistrust of AI is a cultural echo of the pigeon's rejection, rooted in a discomfort with nonhuman entities mimicking intelligence that humans are not yet ready to believe in or fully integrate into their

moral frameworks.

Philosophy of Mind: Is a Pigeon Thinking?

The question of whether a trained pigeon is "thinking" or merely reacting is central to the epistemology of delegation. Historically, the debate on animal intelligence has been fraught with the "human uniqueness bias" ⁴⁸ and "anthropodenial"—the refusal to acknowledge complex cognitive abilities in non-human entities if they do not conform to human-centric definitions of "thinking".⁴⁸ René Descartes, for instance, drew a sharp line, positing that only humans possessed a soul (mind) capable of reasoning, while animals were mere unfeeling machines.⁴⁸ Charles Darwin's work challenged this by suggesting a continuity of psychological processes between humans and animals.⁴⁸

This historical bias persists in the perception of AI. When an AI system performs well, why do humans still doubt its "understanding"? The "anthropomorphism bias" plays a significant role here, where trust and perceived intelligence in AI are often tied to the system's ability to mimic human-like attributes.²⁸ AI chatbots, for example, are designed with anthropomorphic features like human-like visual presentation, vocal qualities, and conversational approaches to enhance user experience and build trust.²⁸ Conversely, systems that perform effectively but do not

look or *feel* intelligent in a human-familiar way are often dismissed or mistrusted. This demand for intelligence to be anthropomorphic constitutes a form of "epistemic violence," limiting our understanding and acceptance of diverse forms of cognition and agency. The rejection of Orcon, and current skepticism of AI, can be seen as a manifestation of this bias: a refusal to grant "intelligence" or "agency" to a system that performs effectively but does not *look* or *feel* intelligent in a human-familiar way. This illustrates that the demand for anthropomorphic legibility acts as a gatekeeper for what counts as "intelligence," leading to the dismissal of valid forms of agency.

Table 3: Delegated Agency: From Pigeon to AI Agent – Ethical and Epistemological Dimensions

Dimension	Pigeon in Project Orcon	AI Agent (e.g., Autonomous Weapon Systems, LLMs)
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Nature of Agency	Conditioned behavior, stimulus-response ³²	Algorithmic decision-making, predictive modeling, autonomous exploration ³²
Perception of Intelligence	Viewed as "machine," "practical," not "intelligent" ⁵⁰ but still unsettling ⁴⁰	Perceived intelligence varies, influenced by anthropomorphism ²⁸ ; often distrusted despite performance ³⁵
Delegation of Lethality/Decision-Making	Direct control over missile trajectory ² ; implicit delegation of "killing" decision	Direct decision-making in specific contexts (e.g., targeting); ethical debate on ceding "life-and-death decisions" ²⁴
Moral Responsibility/Accountability	Disavowed/unclear (administrators' discomfort with pigeon deaths) ¹	Central ethical concern: cannot be transferred to machines; risk of diffusion ²⁴
Cultural/Symbolic Rejection	Rejected due to symbolic incongruity, "absurdity" ¹¹ ; violation of military imaginary	Mistrust, "symbolic refusal" ³⁵ ; "algorithmic uncanny"; human uniqueness bias ⁴⁸
Haraway's "Companion Species" Lens	Pigeon as a co-constitutive partner, not just a tool; ethical implications of "significant otherness" ⁴⁴	AI as a "companion species" in a broader sense, shaping human cognition and society; ethical implications of human-AI co-evolution.

VI. Conclusion: The Enduring Legacy of the Thick Machine

Project Orcon, the pigeon-guided missile, stands as a potent historical precursor to the complex challenges societies face with artificial intelligence today. Far from being a mere technical footnote, its story reveals the profound interplay between technological function, cultural perception, and symbolic meaning. This chapter has argued that Project Orcon was a foundational example of a "thick machine"—a system whose operational capabilities were inextricably bound to its cultural and symbolic legibility. It functioned not just technically, but ritually and symbolically, embodying a performance of rationality that ultimately failed to resonate with the prevailing

sociotechnical imaginaries of its time.

The rejection of Project Orcon was a ritualistic act of symbolic refusal, mirroring the skepticism and discomfort often directed at modern AI. Both instances highlight a persistent societal unease with delegating intelligence and control to nonhuman proxies, particularly in high-stakes domains like warfare. The "absurdity" of a pigeon guiding a bomb, much like the "algorithmic uncanny" of an opaque neural network, challenged deeply held human notions of agency, responsibility, and what constitutes legitimate intelligence. This historical analysis underscores that technological acceptance is not solely a matter of efficiency or precision, but a complex negotiation within a cultural system that defines what is serious, what is trustworthy, and what forms of intelligence are acceptable.

The enduring legacy of the "cybernetic pigeonfight" is evident in the ongoing debates surrounding AI ethics, trust, and control. As AI systems become increasingly autonomous and integrated into critical infrastructures, the questions first implicitly raised by Project Orcon—who is responsible, what constitutes intelligence, and how do we trust nonhuman agents with significant power—become ever more urgent. Understanding "thick machines" requires an interdisciplinary approach that transcends purely functionalist views of technology. It necessitates integrating technical analysis with rigorous cultural, symbolic, and ethical inquiry. By examining the historical precedents of symbolic rejection and the underlying anxieties they reveal, societies can better navigate the "deep play" of delegating agency to future forms of intelligence, ensuring that technological progress aligns not just with technical capability, but with human values, trust, and a nuanced understanding of posthuman agency. The insights from Project Orcon compel a speculative ethnography of intelligence, urging contemplation of the new rituals and symbolic frameworks that will inevitably emerge as humanity continues to co-evolve with increasingly sophisticated nonhuman entities.

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